

1.

```
class Student
{
    string name;
};
class Maths
{
    Student student1;
};
```

This code represents:

- A. An 'is a' relationship**
- B. A 'has a' relationship**
- C. Both**
- D. Neither**

2. What is the error in the following code:

```
class Test
{
    virtual void Display();
};
int main()
{
    return 0;
}
```

- A. No error**
- B. Function Display() should be declared as static**
- C. Function Display () should be defined.**
- D. Test class should contain data members.**

3. class Person

```
{ };
```

```
class Student : protected Person
```

```
{ };
```

```
int main()
```

```
{
    return 0;
}
```

What will happen when we try to compile this code?

- A. Will not compile because class body of person is not defined.**
- B. Will not compile because the class body of student is not defined.**
- C. Will not compile because class Person is not public inherited.**
- D. Will compile successfully**

4.

```
class A
{
public:
    virtual void show() = 0;
};
class B :public A
{
public:
    void display()
    {
        cout << "B" << endl; }

};
class C : public A
{
public:
    void show()
    {
        cout << "C" << endl; }

};
```

Which of the following statement is/are illegal?

- A. C c1;**
- B. A a1;**
- C. B b1;**
- D. A *arr[2];**

5.

```
class Base
{private: int x;
 protected: int y;
};
class Derived : public Base
{    int a, b;
    void change()
    {        a = x;
             b = y;  }
};
int main()
{    Base base;
    Derived derived;
    base.y = 0;
    derived.y = 0;
    derived.change();
    system("pause");
    return 0;}
```

which of the following line/lines in the above program will produce compilation errors?

- A. a=x;**
- B. b=y;**
- C. base.y;**
- D. derived.y;**

```

6. class INHA
{private:
    int a;
public:
    void INHA(){ a = 0; cout << "Object created"; }
    void INHA(int a){ x = a; }
};
int main()
{
    INHA i;
        return 0; }

```

```

7. class Myclass
{
    private:
    char data;
};
class Myderived : public Myclass
{private:
    char data1;
};
int main()
{
    Myclass b;
    Myderived d1;
    cout << endl << sizeof(b) << endl << sizeof(d1) << endl;
        return 0;

}

```

8.

```
class OOP
{public:

virtual~OOP()
{
cout << endl << "In OOP  destructor" << endl;
}
};
class OOP1 : public OOP
{
public:
~OOP1()
{      cout << endl << "In OOP1 destructor" << endl;      }
};
int main()
{
OOP *b;
    b = new OOP1;
    delete b;
    return 0;}
```

```

9. class Base
{public:
    virtual void show()
    {cout << " In Base \n"; }
};
class Derived : public Base
{public:
    void show()
    {      cout << "In Derived \n";      }
};
int main(void)
{
    Base *bp = new Derived;
    bp->show();
    Base &br = *bp;
    br.show();
    bp->show();
    (*bp).show();
    return 0;}

```

```

10. int main()
{
    int x = 'C';
    fstream outfile("Ram.txt",ios::out);
    outfile<<x;
    outfile.close();
    ifstream infile("Ram.txt",ios::in);
    infile >> x;
    cout << "\nvalue= " << x << endl;
    return 0;
}

```

```
11. class Result
{private:
int m;
public:
void getdata(){
    cout << "Enter Number:";
    cin >> m;    }
    void display(){ cout << m; }    //M=2
};
int main(){
    Result R;
    R->getdata();
    R->display();
    Result *p = new Result;
    p.getdata();
    (*p).display();
    return 0;
}
```

```

12. class A
{protected:
    int a, b;
public:A(int y, int x=0)
    {
        a = x;
        b = y; }

    A() { a = b = 0; }
    virtual void print();
};

class B : public A
{private:
    float p, q;
public: B(int m, int n, float u, float v) : A(m,n)
    {
        p = u;
        q = v; }
    B(){ p = q = 0; }
    void input(float u, float v);
    virtual void print(float);
};

void A::print(void)
{
    cout << "A values: " << a << " " << b << endl;}
void B::print(float)
{
    cout << "B values: " << p << " " << q << endl;}
void B::input(float x, float y)
{p = x;    q = y;}
int main()
{
    A a1(10, 20), *ptr;
    B b1;
    b1.input(7.5, 3.14);
    ptr = &a1;
    ptr->print();
    return 0;
}

```

```
13. class B1
{public:
void display()
{cout <<"B1" << endl;}
};
class B2
{public:
void display()
{cout << "B2" << endl;}
};
class D:public B1,public B2
{
//nothing here
};
int main()
{
    D d;
    d.display();
    d.B1::display();
    d.B2::display();
    return 0;
}
```